

RAMDANI KHALED

AI Systems Builder | Analytics Engineer

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PROFESSIONAL SUMMARY

Final-year Computer Science student at Université Badji Mokhtar – Annaba, specializing in Analytics Engineering and AI. Currently building AI-powered voice assistants and multi-agent ML systems. Machine learning builder driven by one goal: build AI systems with real-world impact.

EXPERIENCE

Founding AI Engineer

Jan 2026 – Present

Avanova – AI-Powered Business Assistants | Annaba, Algeria

- Built the full AI pipeline for the Avanova Smart Avatar Assistant — speech recognition, knowledge retrieval, language generation, and voice synthesis — with real-time streaming
- Built a 3D avatar system with real-time lip sync driven by live audio analysis across custom facial morph targets
- Achieved sub-1-second end-to-end response latency through GPU acceleration — down from 10+ seconds on CPU
- Implemented multilingual support (English, French, Arabic) with automatic language detection and per-language voice switching
- Designed the full system architecture: backend, vector database with per-client isolation, automatic voice detection, error handling, and auto-reconnect
- Delivered a production-ready B2B MVP — a voice avatar assistant for hotel reception desks, retail entrances, and office environments — in under 2 months, solo
- Demonstrated the system live at a public launch event in front of a real audience

AI Systems Intern

Jan 2026 – Present

ANEM (Agence Nationale de l'Emploi) – Algerian National Employment Agency

- Building an employment intelligence platform using a multi-agent ML architecture on large-scale data
- Designed a four-stream data pipeline spanning Profiles, Offers, Placements, and References
- Built a recommendation engine for skill gap analysis and job-candidate matching
- Implemented dynamic weight optimization using supervised learning trained on historical placement data

PROJECTS

Avanova Smart Avatar Assistant

2026

- Full-stack AI voice assistant with 3D avatar: speaks, listens, and responds in real time using grounded, context-aware responses
- Real-time lip sync via live audio analysis driving facial animation at 30 FPS
- Sub-1-second voice response on GPU; fully offline after first setup; trilingual (EN/FR/AR)
- Deployed as a B2B SaaS MVP for hotel lobbies, retail entrances, and office receptions

ARIA – Adaptive Reasoning & Intelligence Architecture

In Development

- Multi-brain AI agent (Planner, Advisor, Meta-Evaluator) with human approval checkpoint before execution
- Three-layer memory system combining working, episodic, and strategic knowledge graph memory
- Privacy-first local inference with optional cloud fallback and cryptographic audit trails per step

AI Engine – Employability Scoring System

Active · Internship

- Multi-agent ML system for employment analytics processing large-scale workforce records
- Per-strategy models with feature importances fed back as dynamic weight overrides

Wyag Lite – Git Reimplementation in Python

Complete

- Rebuilt Git internals from scratch in pure Python with zero external dependencies; implements commits, trees, refs, tags, and log

TECHNICAL SKILLS

AI & Agents: Multi-agent architecture, LLM orchestration, RAG, vector stores, local inference, agentic pipelines, real-time voice AI

Voice AI: Speech recognition, text-to-speech, voice activity detection, real-time streaming, lip sync, avatar animation

Machine Learning: scikit-learn, NumPy, Pandas, OpenCV, MediaPipe, supervised learning, KNN

Data Engineering: MongoDB, Elasticsearch, vector databases, NoSQL design, SQL

GPU & Inference: CUDA, GPU-accelerated inference, performance optimization

Development: Python (primary), JavaScript, React, REST APIs, Git, C, Java (basic)

EDUCATION

Bachelor of Science in Computer Science

Expected: 2026

Université Badji Mokhtar – Annaba, Algeria | Member, AI & Robotics Association of Annaba Wilaya

LANGUAGES & AFFILIATIONS

Arabic (Native) | English B1-B2 (3 years university instruction) | Member, AI & Robotics Association of Annaba Wilaya (government-recognized)